

# Amulet Motion Controller

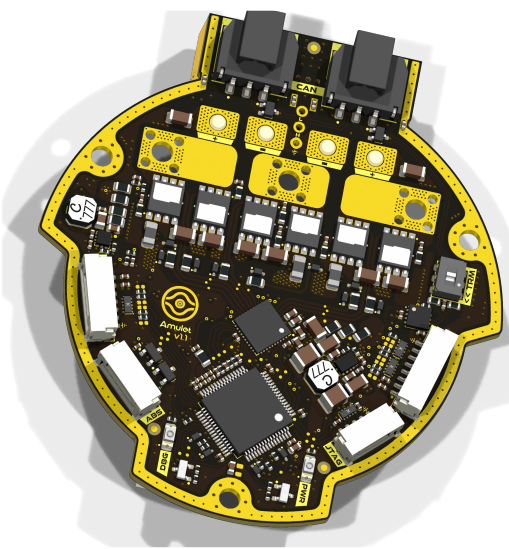
Variant:

2026-08-17

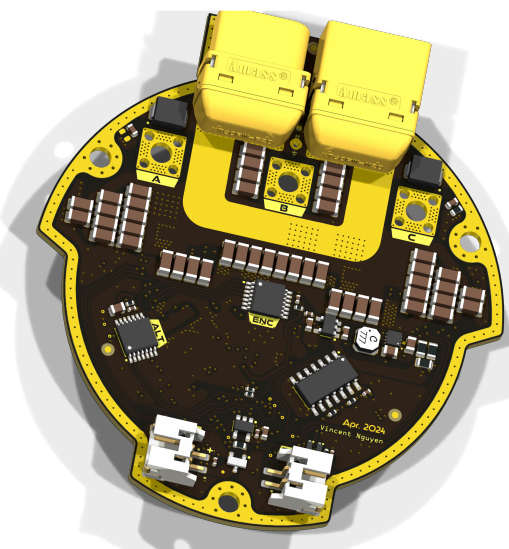
Rev 1.1.1+ (Unreleased)

| Page | Index                     | Page | Index                     | Page | Index            | Page | Index |
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TOP VIEW



BOTTOM VIEW



## NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 17-Aug-2026

## DESIGN CONSIDERATIONS

DESIGN NOTE:  
Example text for informational design notes.

DESIGN NOTE:  
Example text for debug notes.

DESIGN NOTE:  
Example text for cautionary design notes.

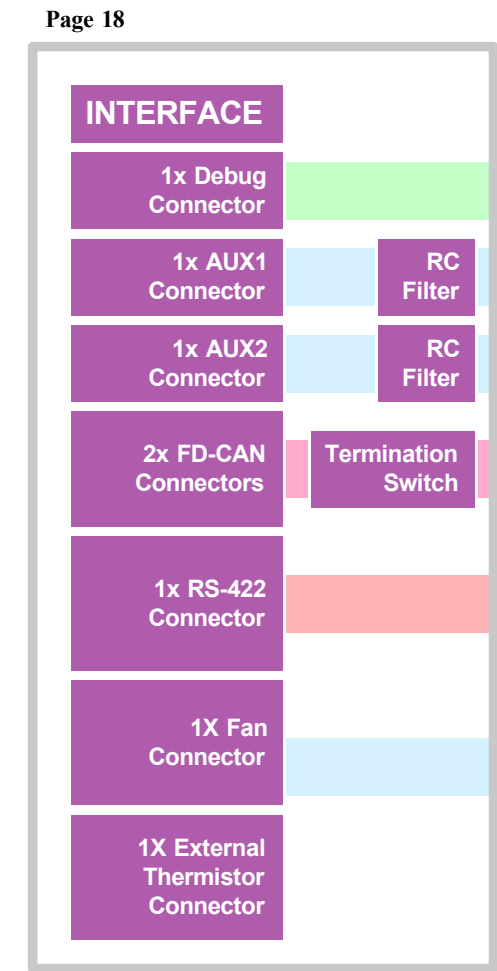
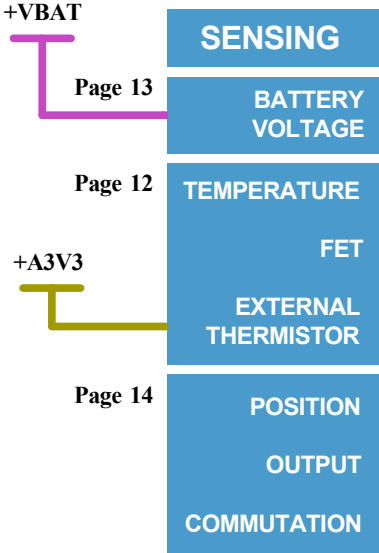
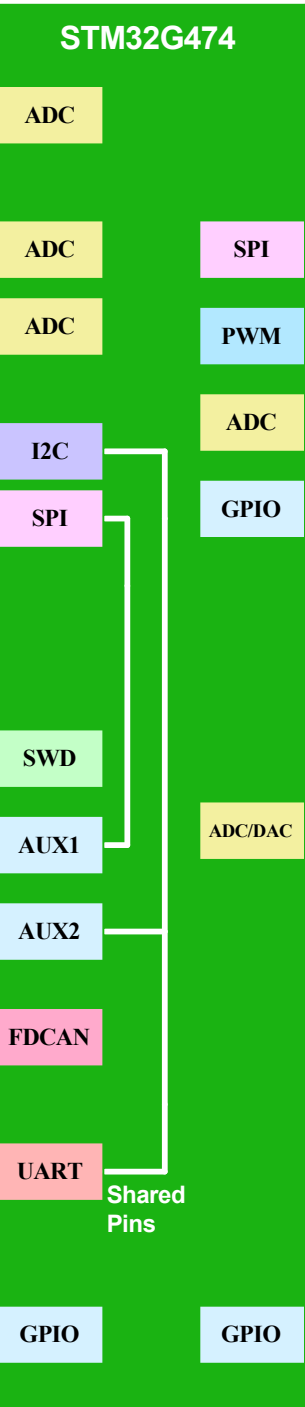
DESIGN NOTE:  
Example text for critical design notes.

LAYOUT NOTE:  
Example text for critical layout guidelines.

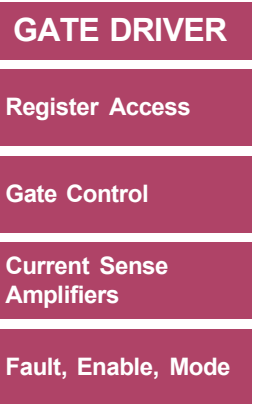
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|  | Comments:                  | Company:<br>EPFL Xplore Research          | Variant:                         | Git Hash:<br>5b68472 |
|  | Sheet Title:<br>Cover Page | Board Name:<br>Amulet Motion Controller   | Project Name:<br>Chienpanzé      |                      |
|  | Sheet Path:<br>/           | File Name:<br>amulet_controller.kicad_sch | Designer:<br>Vincent Nguyen      | Date:<br>2024-04-13  |
|  |                            |                                           | Revision:<br>1.1.1+ (Unreleased) |                      |
|  |                            |                                           | Size:<br>A3                      | Sheet:<br>1 of 21    |

# [2] Block Diagram

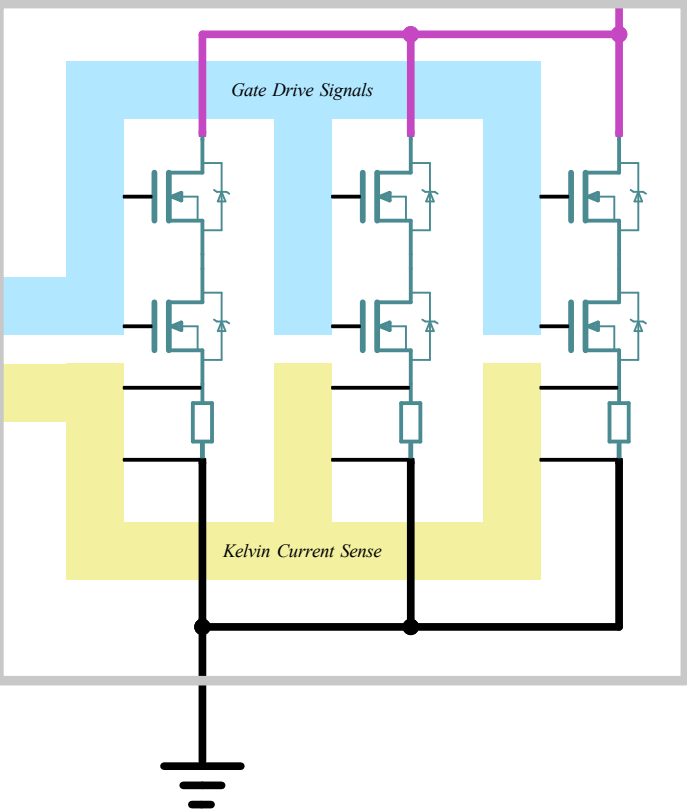
Page 4 (Power)  
Page 5 (I/Os)



Page 8



Page 9



Page 10

**BOARD  
VERSION,  
DAC DEBUG**

Page 11

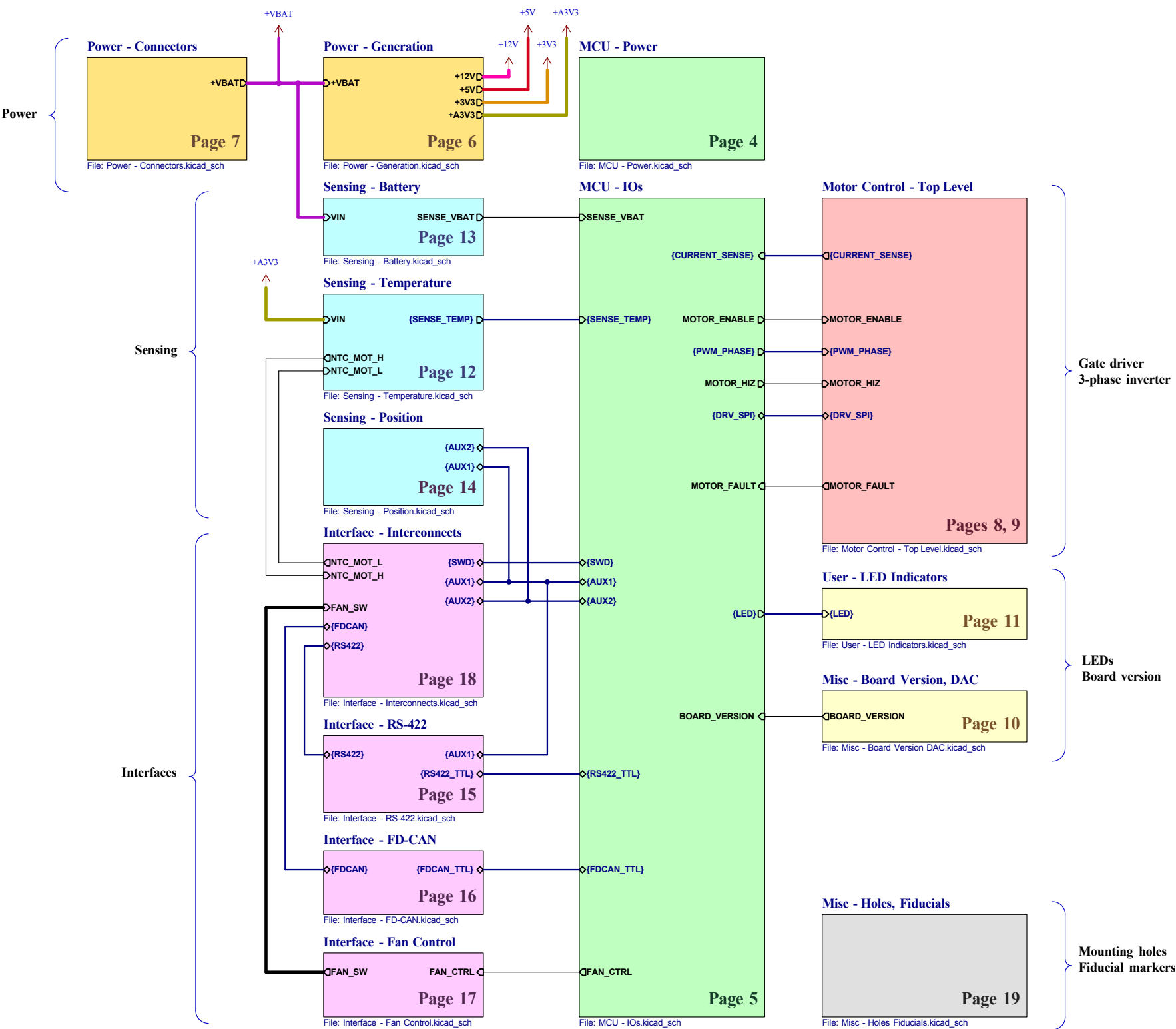
**2x USER LEDs**  
(GREEN, YELLOW)

**Target specifications:**

|                      |           |
|----------------------|-----------|
| Input voltage:       | 12-44 V   |
| Continuous current:  | 30 A      |
| Peak current:        | 100 A     |
| Switching frequency: | 15-60 kHz |

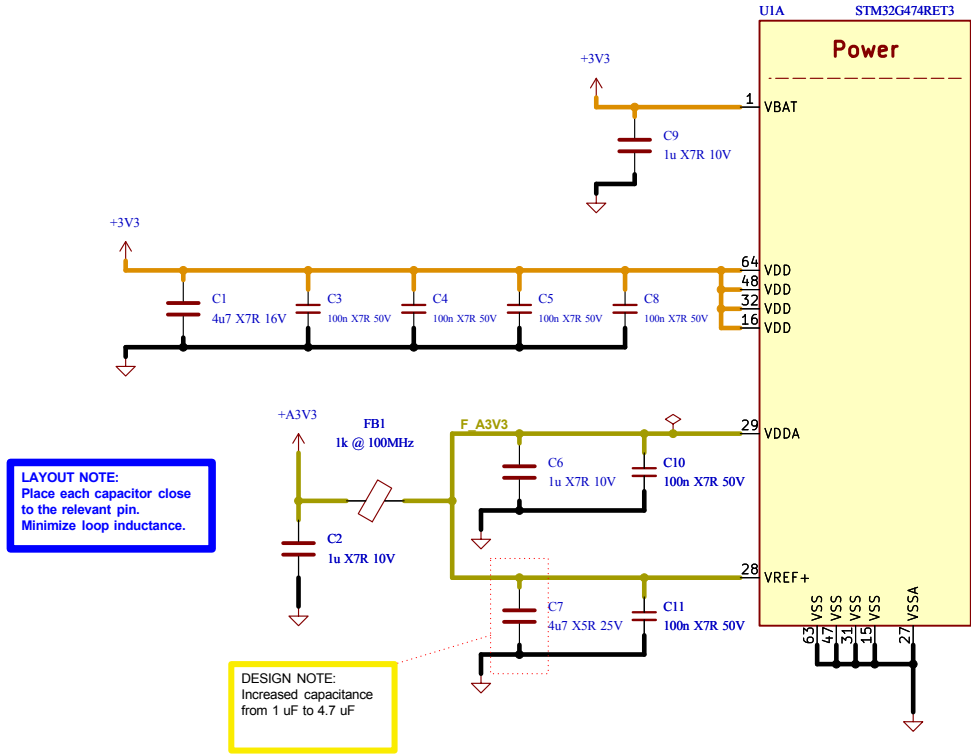
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| Comments:                      | Company:<br>EPFL Xplore Research               |                             | Variant:                           | Git Hash:<br>5b68472             |
|                                | Board Name:<br><b>Amulet Motion Controller</b> |                             | Project Name:<br><b>Chienpanzé</b> |                                  |
| Sheet Title:<br>Block Diagram  | File Name:<br>Block Diagram.kicad_sch          | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13                | Revision:<br>1.1.1+ (Unreleased) |
| Sheet Path:<br>/Block Diagram/ |                                                | Reviewer:                   | Size:<br><b>A3</b>                 | Sheet:<br><b>2</b> of <b>21</b>  |

# [3] Project Architecture



|                                       |                                              |                             |                             |                                  |
|---------------------------------------|----------------------------------------------|-----------------------------|-----------------------------|----------------------------------|
| Comments:                             | Company:<br>EPFL Xplore Research             |                             | Variant:                    | Git Hash:<br>5b68472             |
|                                       | Board Name:<br>Amulet Motion Controller      |                             | Project Name:<br>Chienpanzé |                                  |
| Sheet Title:<br>Project Architecture  | File Name:<br>Project Architecture.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-22         | Revision:<br>1.1.1+ (Unreleased) |
| Sheet Path:<br>/Project Architecture/ |                                              | Reviewer:                   | Size:<br>A3                 | Sheet:<br>3 of 21                |

# [4] MCU - Power

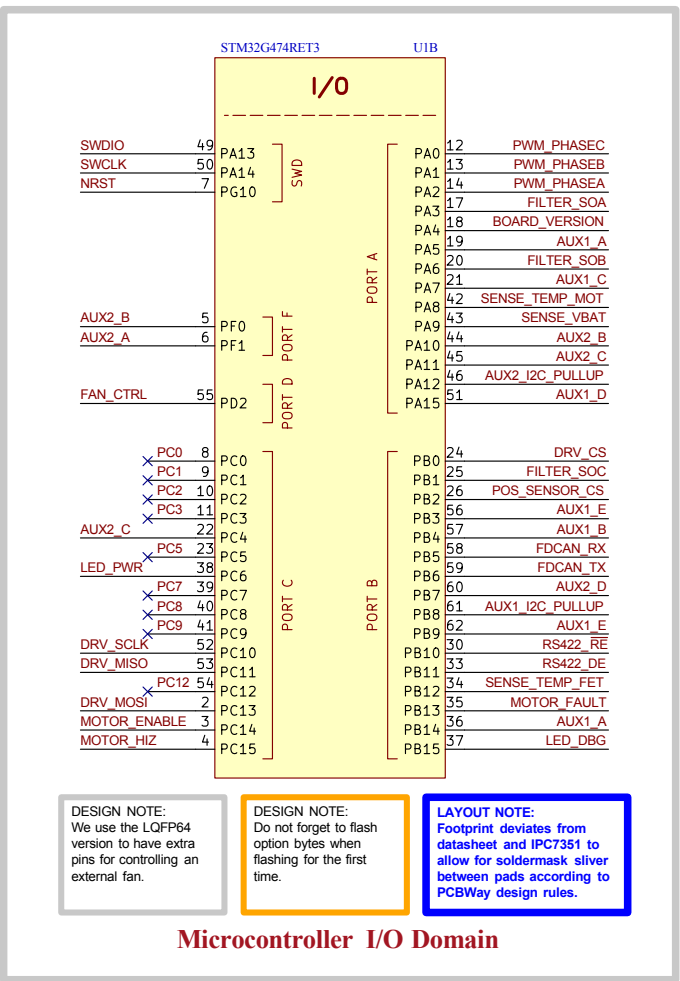


Microcontroller Power Supply Scheme

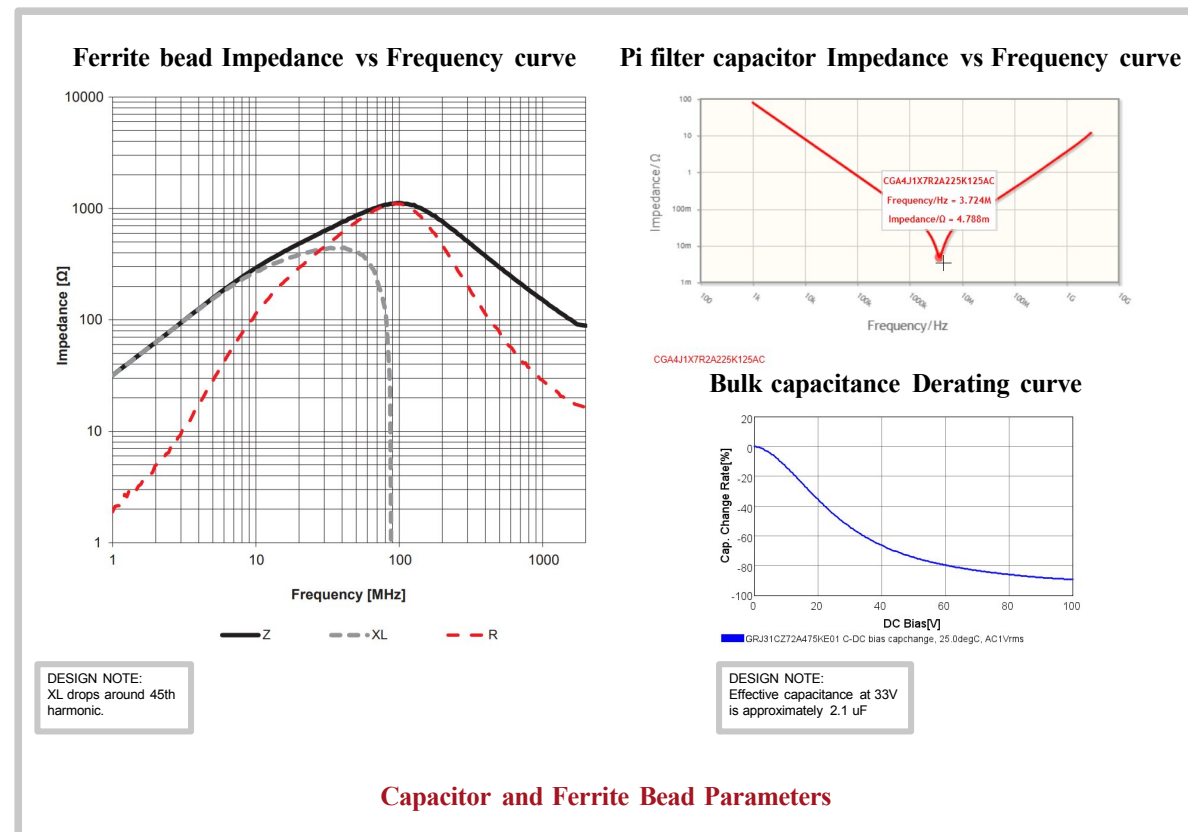
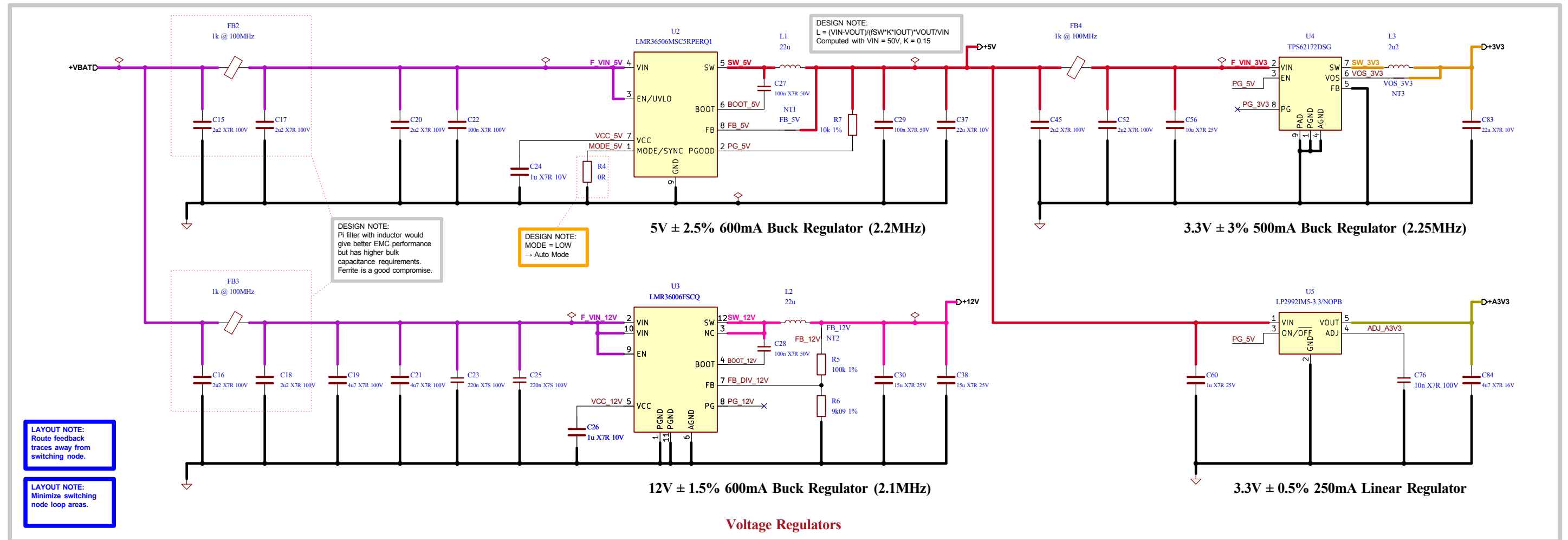
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|  | <b>Comments:</b><br>AN5346<br>STM32G474 Datasheet p.81<br>J. Pieper ADC investigation | <b>Company:</b><br>EPFL Xplore Research        |                                    | <b>Variant:</b>                    | <b>Git Hash:</b><br>5b68472             |
|  | <b>Sheet Title:</b><br>MCU - Power                                                    | <b>Board Name:</b><br>Amulet Motion Controller |                                    | <b>Project Name:</b><br>Chienpanzé |                                         |
|  | <b>Sheet Path:</b><br>/Project Architecture/MCU - Power/                              | <b>File Name:</b><br>MCU - Power.kicad_sch     | <b>Designer:</b><br>Vincent Nguyen | <b>Date:</b><br>2023-12-18         | <b>Revision:</b><br>1.1.1+ (Unreleased) |
|  |                                                                                       |                                                | <b>Reviewer:</b>                   | <b>Size:</b><br>A4                 | <b>Sheet:</b><br>4 of 21                |



# [5] MCU - I/Os



# [6] Power - Generation



**Battery**

- +VBAT TP39 +VBAT
- +VBAT TP40 +VBAT
- +VBAT TP41 +VBAT

**Filtered Power Rails**

- F\_VIN\_12V TP42 F\_VIN\_12V
- F\_VIN\_5V TP43 F\_VIN\_5V
- F\_VIN\_3V3 TP44 F\_VIN\_3V3

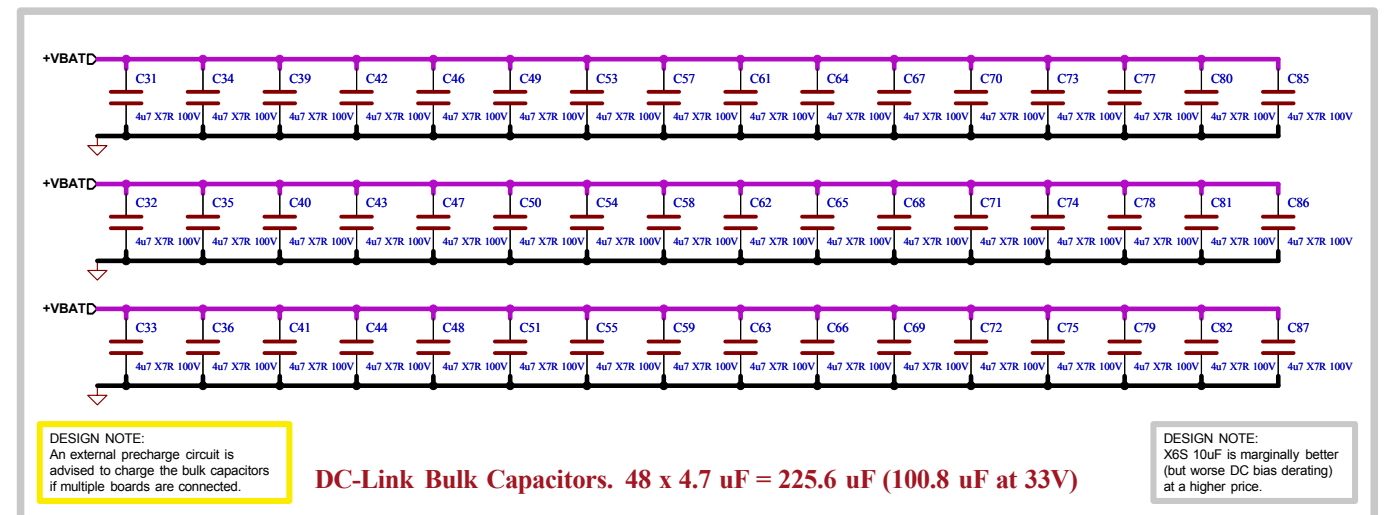
**Power Rails**

- +12V TP45 +12V
- +5V TP46 +5V
- +5V TP47 +5V
- +3V3 TP48 +3V3
- +3V3 TP49 +3V3
- +3V3 TP50 +3V3
- +A3V3 TP51 +A3V3

**Ground**

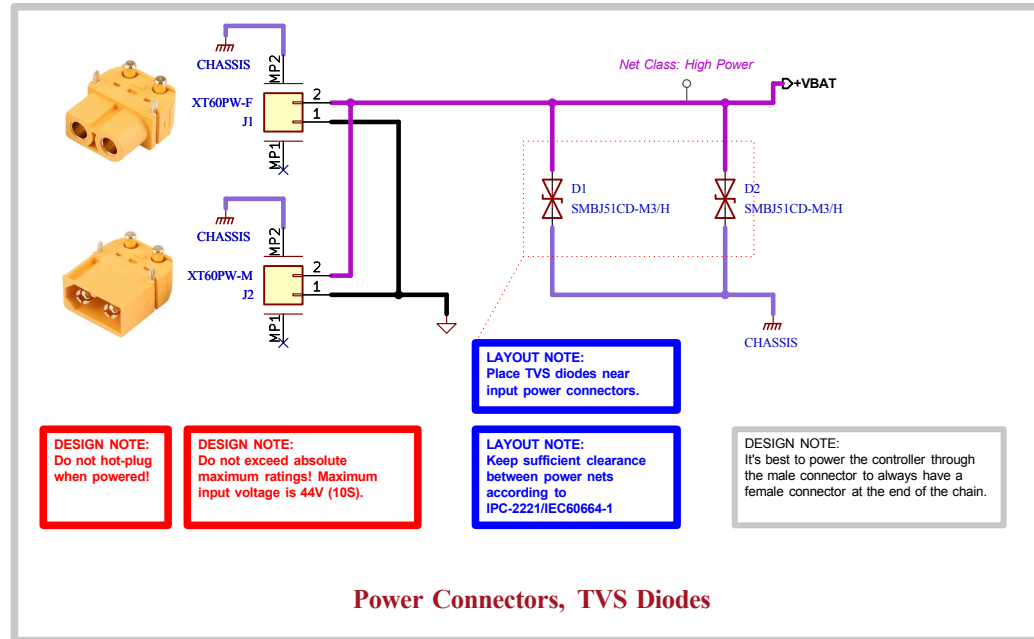
- GND TP52 GND
- GND TP53 GND
- GND TP54 GND
- GND TP55 GND
- GND TP56 GND
- GND TP57 GND
- GND TP58 GND

**Test Points**



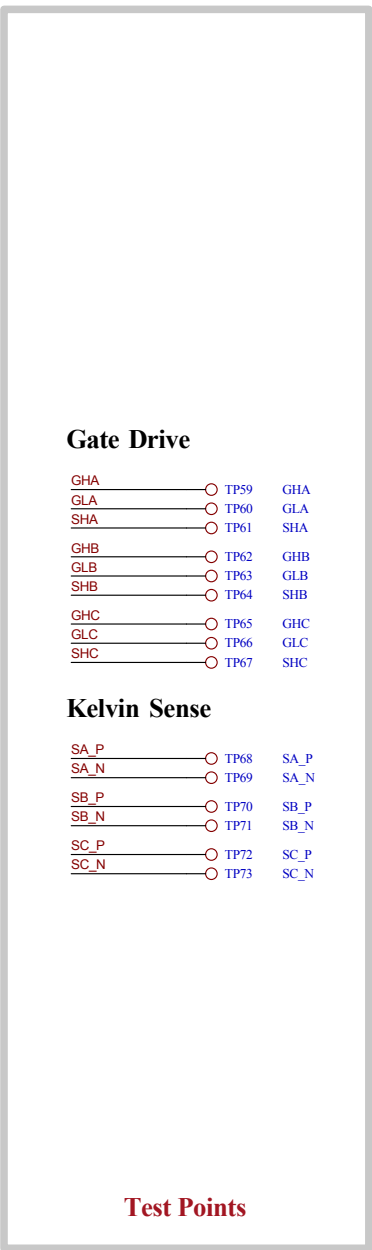
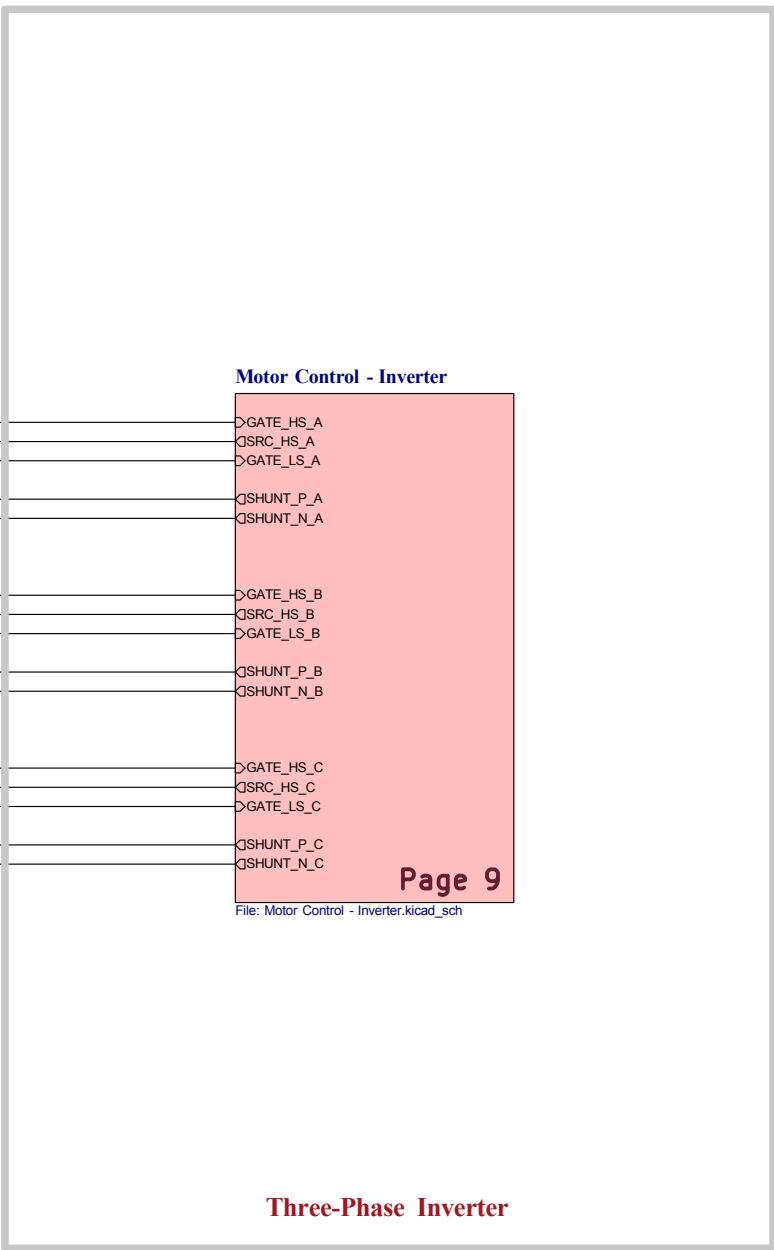
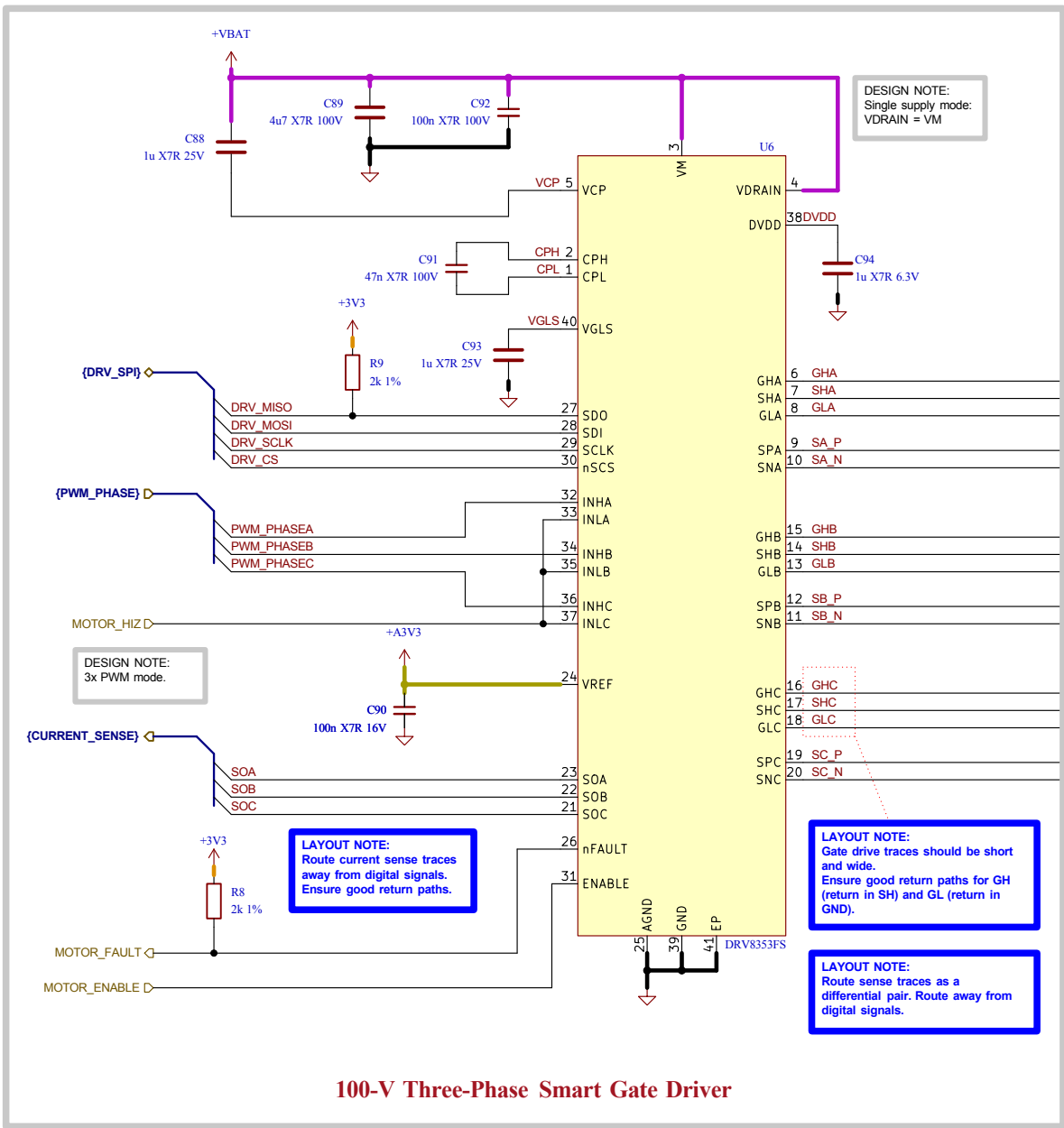
|                                                                                                                                        |  |                                            |                             |                             |                                  |
|----------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------|-----------------------------|-----------------------------|----------------------------------|
| Comments:<br>+12V (600 mA): Fan power<br>+5V (600 mA): Gate Driver, LEDs<br>+3V3 (500 mA): digital 3.3V<br>+3V3A (250 mA): analog 3.3V |  | Company:<br>EPFL Xplore Research           |                             | Variant:                    | Git Hash:<br>5b68472             |
| Sheet Title:<br>Power - Generation                                                                                                     |  | Board Name:<br>Amulet Motion Controller    |                             | Project Name:<br>Chienpanzé |                                  |
| Sheet Path:<br>/Project Architecture/Power - Generation/                                                                               |  | File Name:<br>Power - Generation.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13         | Revision:<br>1.1.1+ (Unreleased) |
|                                                                                                                                        |  | Reviewer:                                  | Size:<br>A3                 | Sheet:<br>6 of 21           |                                  |

## [7] Power - Connectors



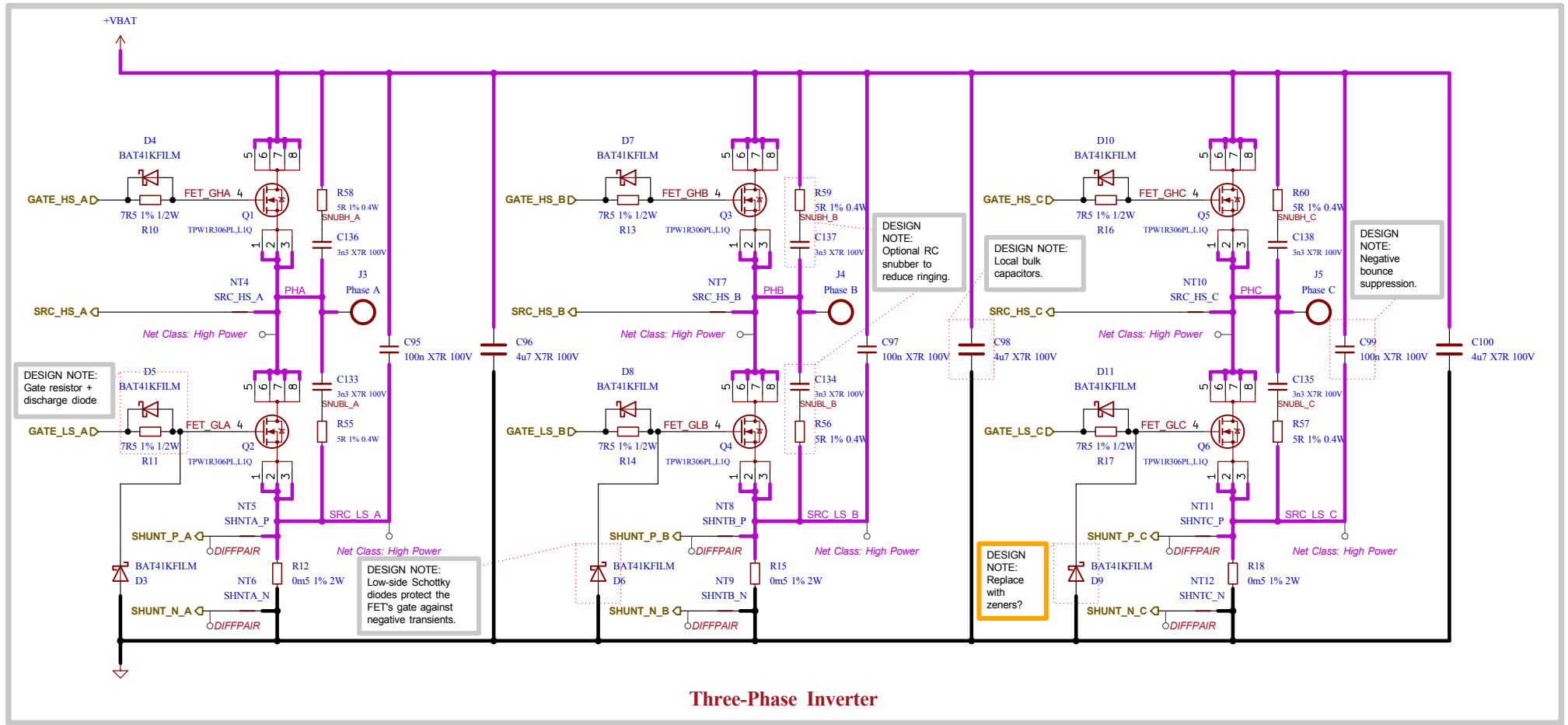
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|  | Comments:                                                | Company:<br>EPFL Xplore Research               | Variant:                           | Git Hash:<br>5b68472             |
|  |                                                          | Board Name:<br><b>Amulet Motion Controller</b> | Project Name:<br><b>Chienpanzé</b> |                                  |
|  | Sheet Title:<br>Power - Connectors                       | File Name:<br>Power - Connectors.kicad_sch     | Designer:<br>Vincent Nguyen        | Date:<br>2023-12-31              |
|  | Sheet Path:<br>/Project Architecture/Power - Connectors/ | Reviewer:                                      | Size:<br><b>A4</b>                 | Revision:<br>1.1.1+ (Unreleased) |
|  |                                                          |                                                |                                    | Sheet:<br><b>7</b> of <b>21</b>  |

# [8] Motor Control - Top Level



|  |                                                                 |                                                   |                                  |                      |
|--|-----------------------------------------------------------------|---------------------------------------------------|----------------------------------|----------------------|
|  | Comments:                                                       | Company:<br>EPFL Xplore Research                  | Variant:                         | Git Hash:<br>5b68472 |
|  | Sheet Title:<br>Motor Control - Top Level                       | Board Name:<br>Amulet Motion Controller           | Project Name:<br>Chienpanzé      |                      |
|  | Sheet Path:<br>/Project Architecture/Motor Control - Top Level/ | File Name:<br>Motor Control - Top Level.kicad_sch | Designer:<br>Vincent Nguyen      | Date:<br>2023-12-20  |
|  |                                                                 |                                                   | Revision:<br>1.1.1+ (Unreleased) |                      |
|  |                                                                 |                                                   | Size:<br>A3                      | Sheet:<br>8 of 21    |

# [9] Motor Control - Inverter



**LAYOUT NOTE:**  
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

**LAYOUT NOTE:**  
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

**DESIGN NOTE:**  
A gate drive current that is too large can damage the FETs!

**Comments:**  
System Design Considerations for High-Power Motor Driver Applications  
Best Practices for Board Layout of Motor Drivers  
Proper RC Snubber Design for Motor Drivers

**Sheet Title:**  
Motor Control - Inverter

**Sheet Path:**  
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

**Company:**  
EPFL Xplore Research

**Board Name:**  
Amulet Motion Controller

**File Name:**  
Motor Control - Inverter.kicad\_sch

**Designer:**  
Vincent Nguyen

**Reviewer:**

**Variant:**

**Git Hash:**  
5b68472

**Project Name:**  
Chienpanzé

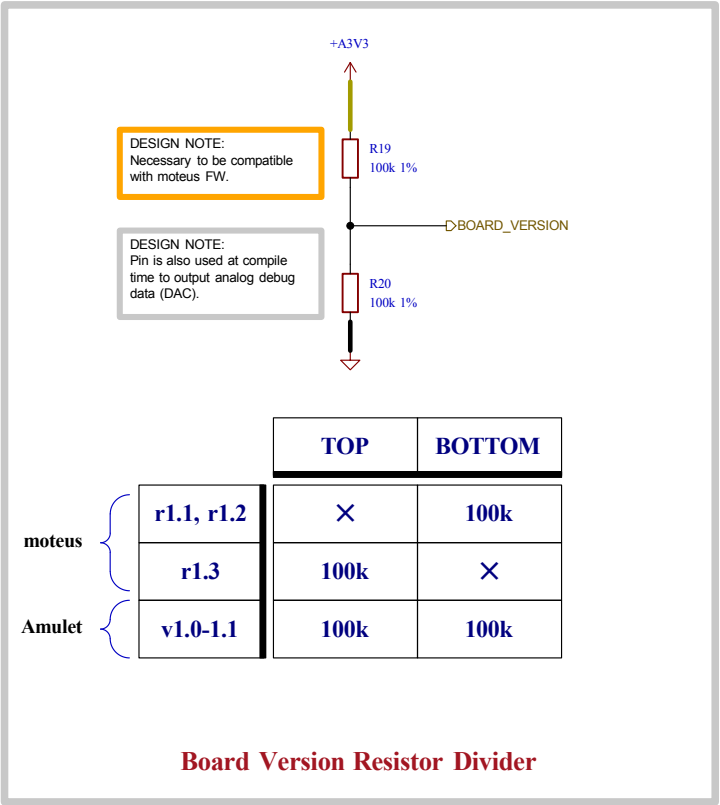
**Date:**  
2024-01-25

**Revision:**  
1.1.1+ (Unreleased)

**Size:**  
A4

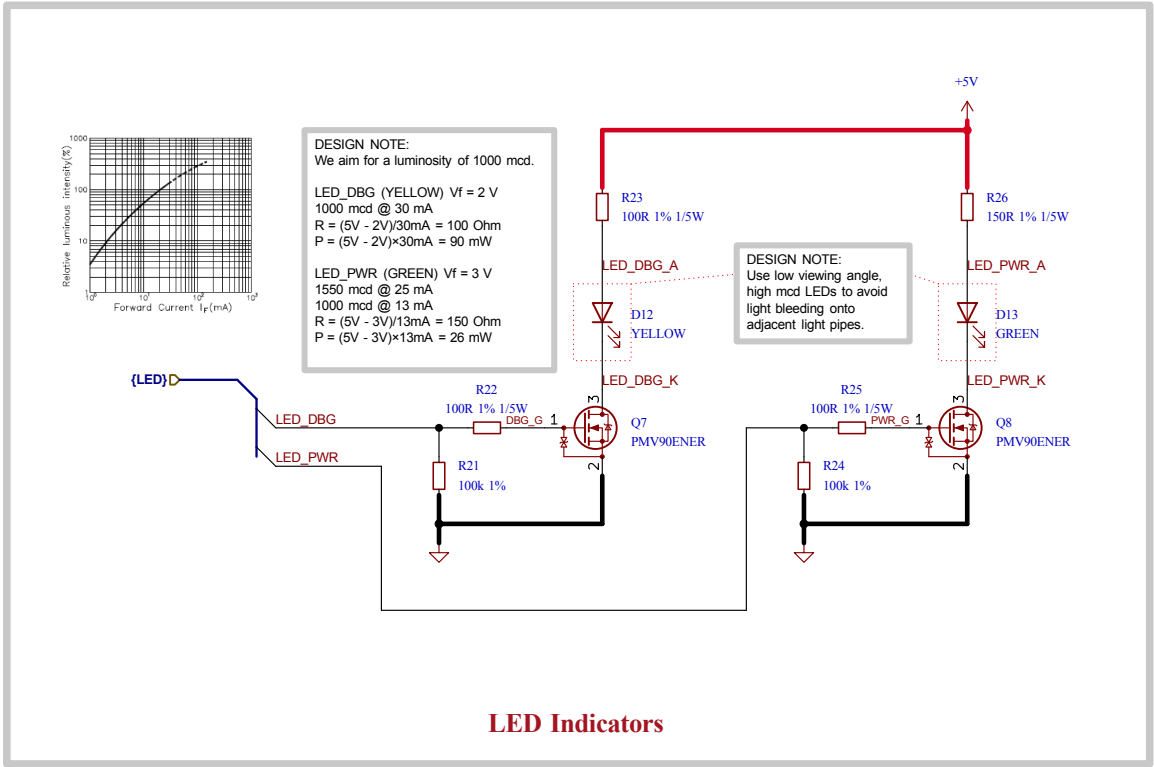
**Sheet:**  
9 of 21

# [10] Misc - Board Version, DAC



|  |                                                                 |                                                  |                             |                             |                                  |
|--|-----------------------------------------------------------------|--------------------------------------------------|-----------------------------|-----------------------------|----------------------------------|
|  | Comments:                                                       | Company:<br>EPFL Xplore Research                 |                             | Variant:                    | Git Hash:<br>5b68472             |
|  |                                                                 | Board Name:<br>Amulet Motion Controller          |                             | Project Name:<br>Chienpanzé |                                  |
|  | Sheet Title:<br>Misc - Board Version, DAC                       | File Name:<br>Misc - Board Version DAC.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13         | Revision:<br>1.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/Misc - Board Version, DAC/ |                                                  | Reviewer:                   | Size:<br>A4                 | Sheet:<br>10 of 21               |

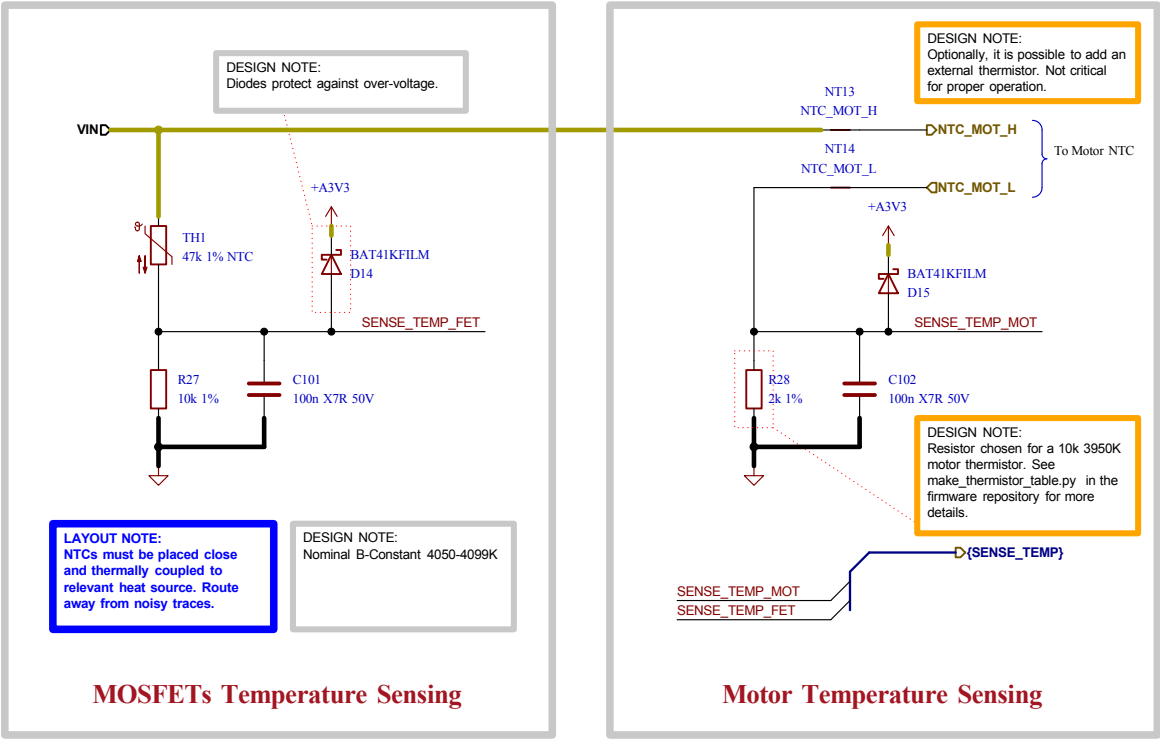
# [11] User - LED Indicators



|  |                                                             |                                               |                             |                             |                                  |
|--|-------------------------------------------------------------|-----------------------------------------------|-----------------------------|-----------------------------|----------------------------------|
|  | Comments:                                                   | Company:<br>EPFL Xplore Research              |                             | Variant:                    | Git Hash:<br>5b68472             |
|  |                                                             | Board Name:<br>Amulet Motion Controller       |                             | Project Name:<br>Chienpanzé |                                  |
|  | Sheet Title:<br>User - LED Indicators                       | File Name:<br>User - LED Indicators.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-19         | Revision:<br>1.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/User - LED Indicators/ |                                               | Reviewer:                   | Size:<br>A4                 | Sheet:<br>11 of 21               |

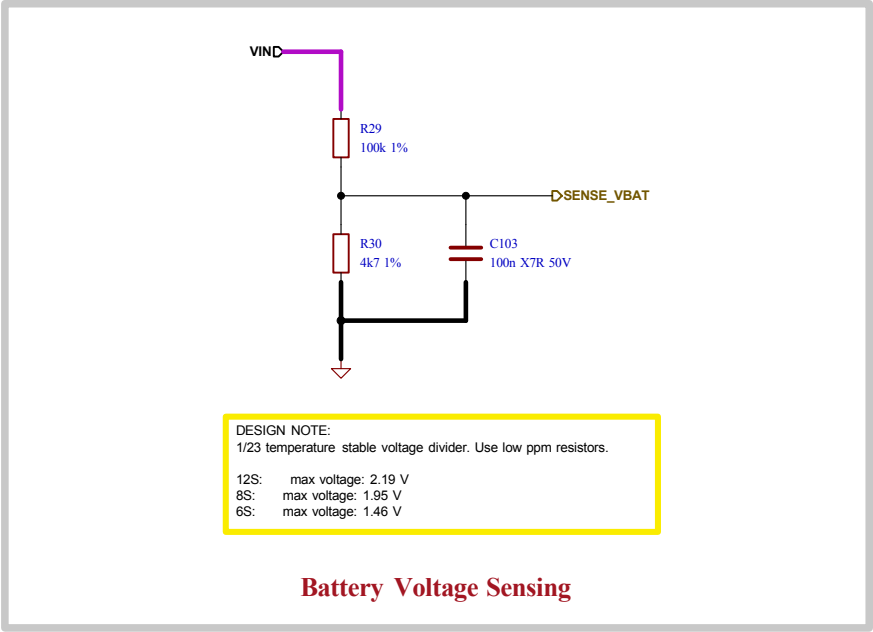


[12] Sensing - Temperature



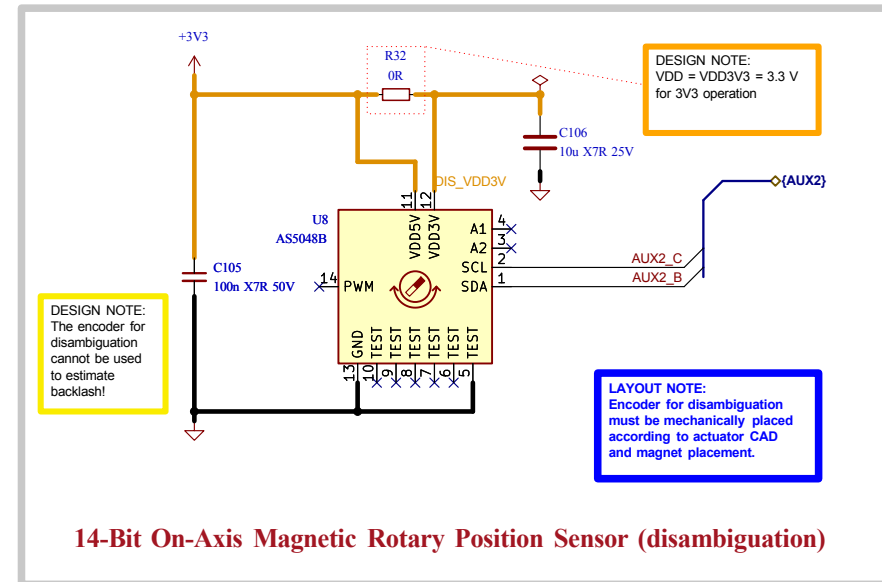
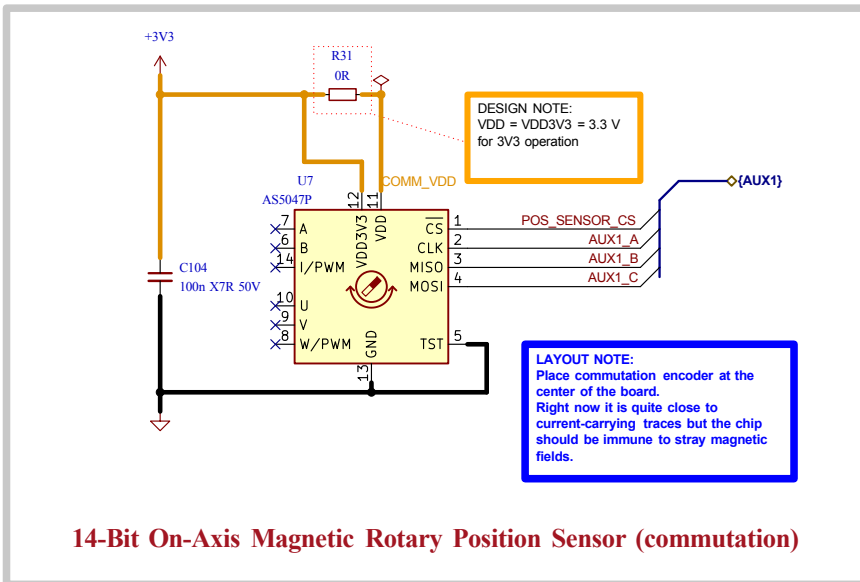
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|  | Comments:                                                   | Company:<br>EPFL Xplore Research              |                             | Variant:                    | Git Hash:<br>5b68472             |
|  |                                                             | Board Name:<br>Amulet Motion Controller       |                             | Project Name:<br>Chienpanzé |                                  |
|  | Sheet Title:<br>Sensing - Temperature                       | File Name:<br>Sensing - Temperature.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13         | Revision:<br>1.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/Sensing - Temperature/ |                                               | Reviewer:                   | Size:<br>A4                 | Sheet:<br>12 of 21               |

# [13] Sensing - Battery



|  |                                                         |                                                |                             |                                    |                                  |
|--|---------------------------------------------------------|------------------------------------------------|-----------------------------|------------------------------------|----------------------------------|
|  | Comments:                                               | Company:<br>EPFL Xplore Research               |                             | Variant:                           | Git Hash:<br>5b68472             |
|  |                                                         | Board Name:<br><b>Amulet Motion Controller</b> |                             | Project Name:<br><b>Chienpanzé</b> |                                  |
|  | Sheet Title:<br>Sensing - Battery                       | File Name:<br>Sensing - Battery.kicad_sch      | Designer:<br>Vincent Nguyen | Date:<br>2023-10-14                | Revision:<br>1.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/Sensing - Battery/ |                                                | Reviewer:                   | Size:<br><b>A4</b>                 | Sheet:<br><b>13</b> of <b>21</b> |

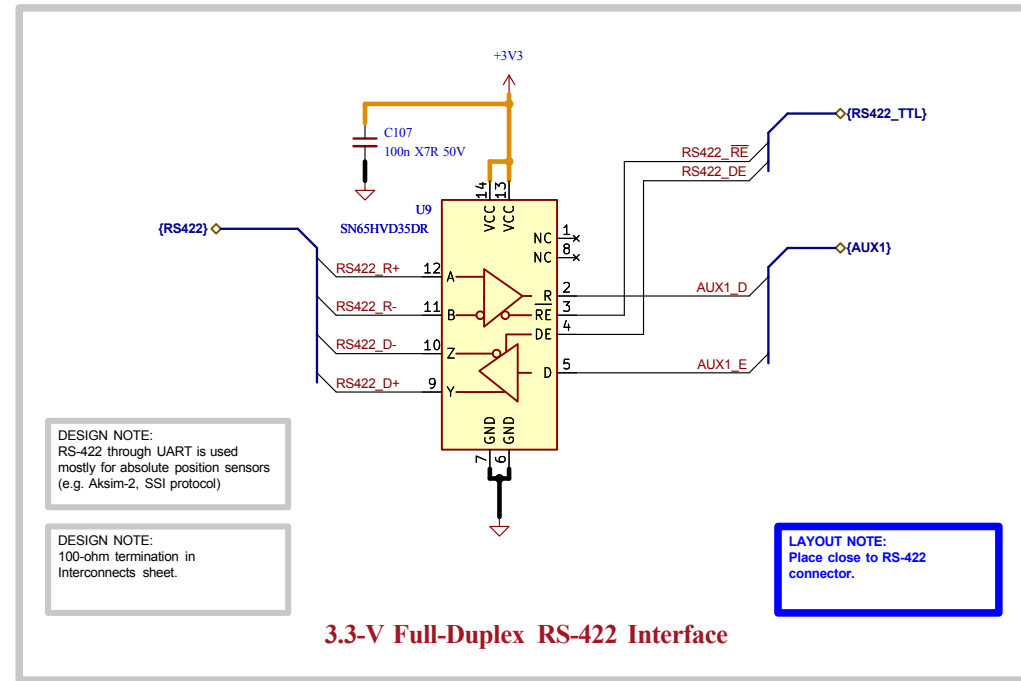
# [14] Sensing - Position



DESIGN NOTE:  
AS5047P senses magnet mounted on planetary sun gear, for commutation.  
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.

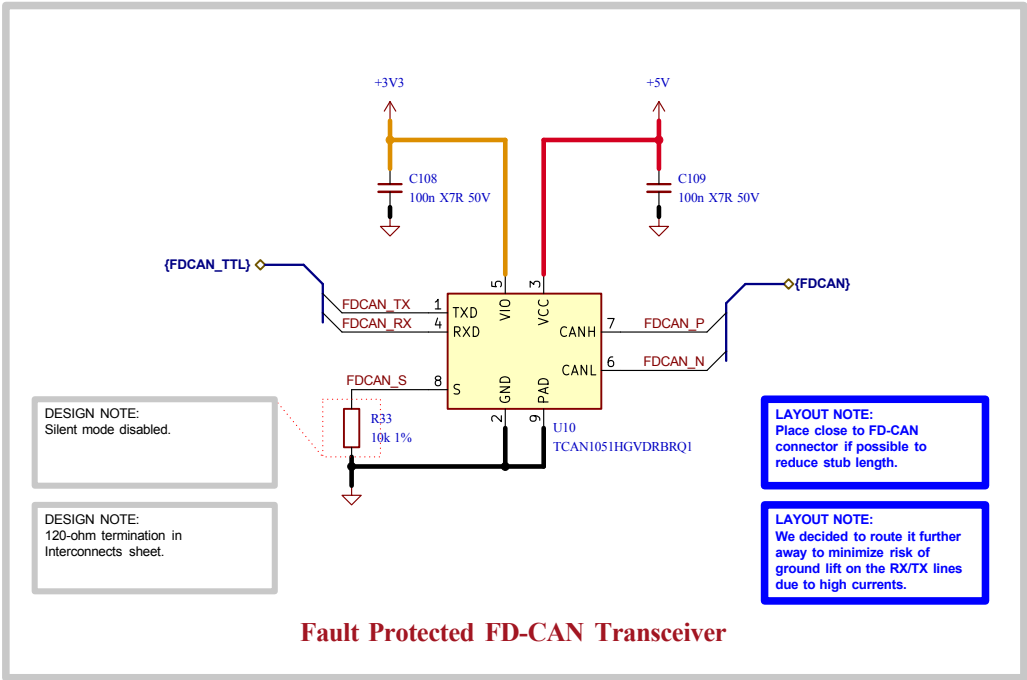
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|  | Sheet Title:<br>Sensing - Position                       | Board Name:<br><b>Amulet Motion Controller</b> | Project Name:<br><b>Chienpanzé</b> |                                  |
|  | Sheet Path:<br>/Project Architecture/Sensing - Position/ | File Name:<br>Sensing - Position.kicad_sch     | Designer:<br>Vincent Nguyen        | Date:<br>2023-10-14              |
|  |                                                          |                                                | Reviewer:                          | Revision:<br>1.1.1+ (Unreleased) |
|  |                                                          |                                                | Size:<br><b>A4</b>                 | Sheet:<br><b>14</b> of <b>21</b> |

# [15] Interface - RS-422



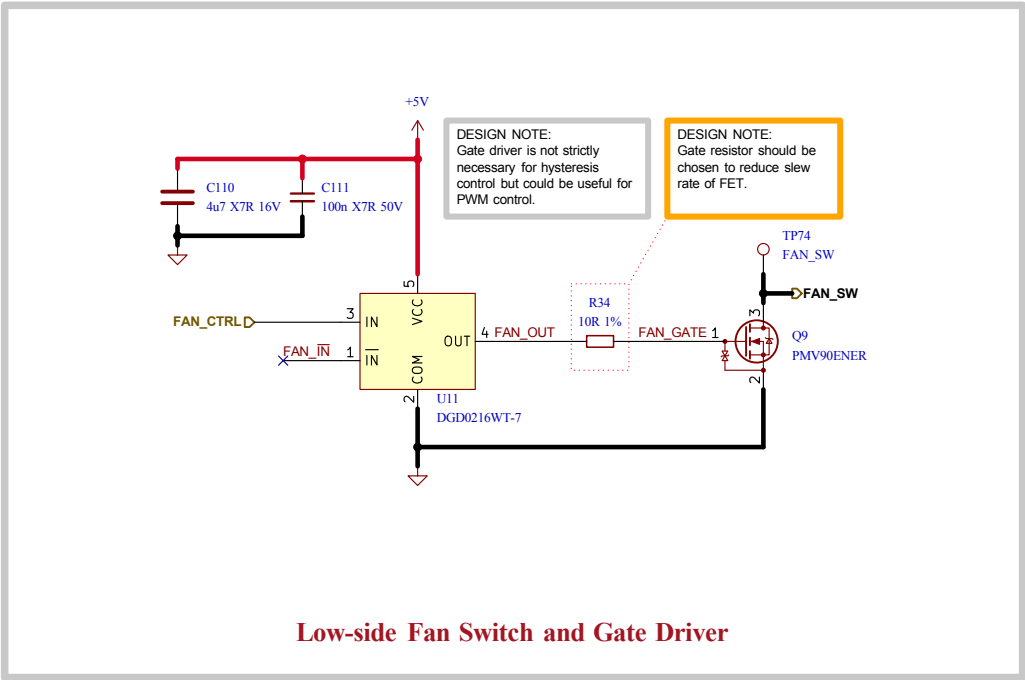
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|  | Comments:                                                | Company:<br>EPFL Xplore Research               |                             | Variant:                           | Git Hash:<br>5b68472             |
|  |                                                          | Board Name:<br><b>Amulet Motion Controller</b> |                             | Project Name:<br><b>Chienpanzé</b> |                                  |
|  | Sheet Title:<br>Interface - RS-422                       | File Name:<br>Interface - RS-422.kicad_sch     | Designer:<br>Vincent Nguyen | Date:<br>2023-10-15                | Revision:<br>1.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/Interface - RS-422/ |                                                | Reviewer:                   | Size:<br><b>A4</b>                 | Sheet:<br><b>15</b> of <b>21</b> |

[16] Interface - FD-CAN



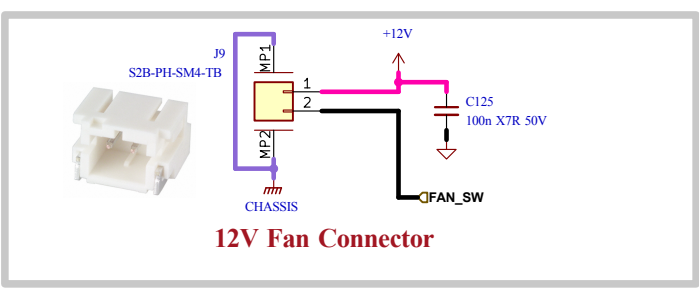
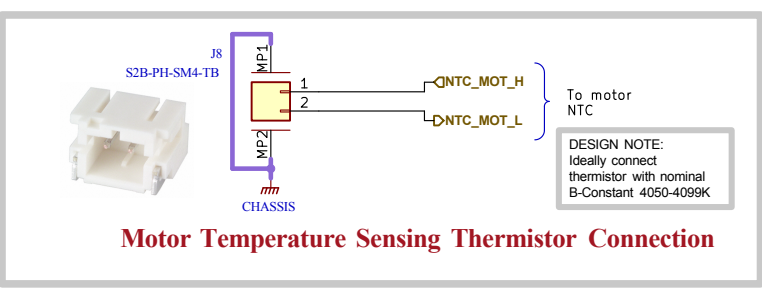
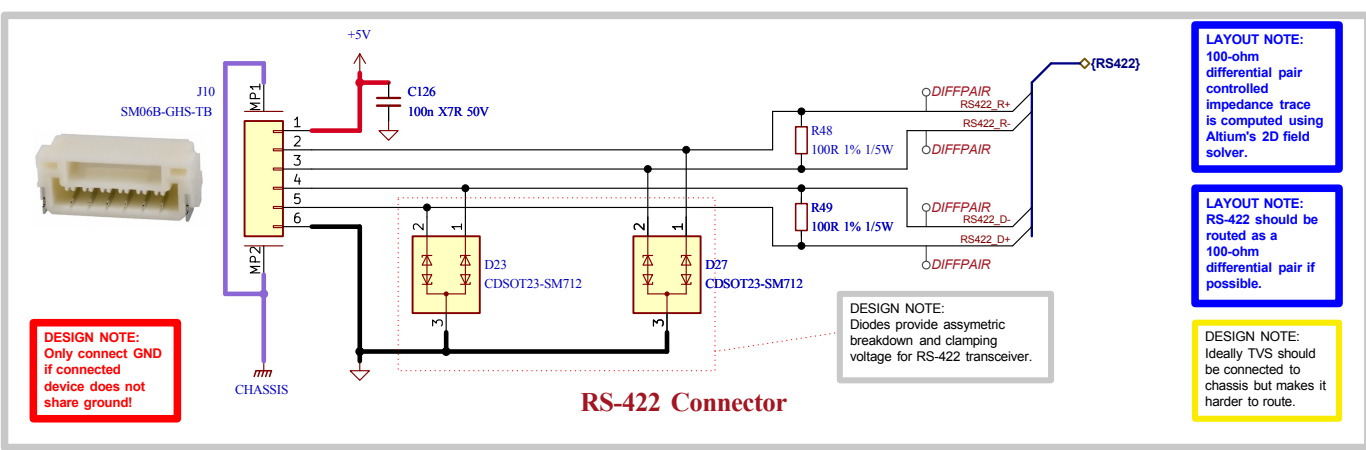
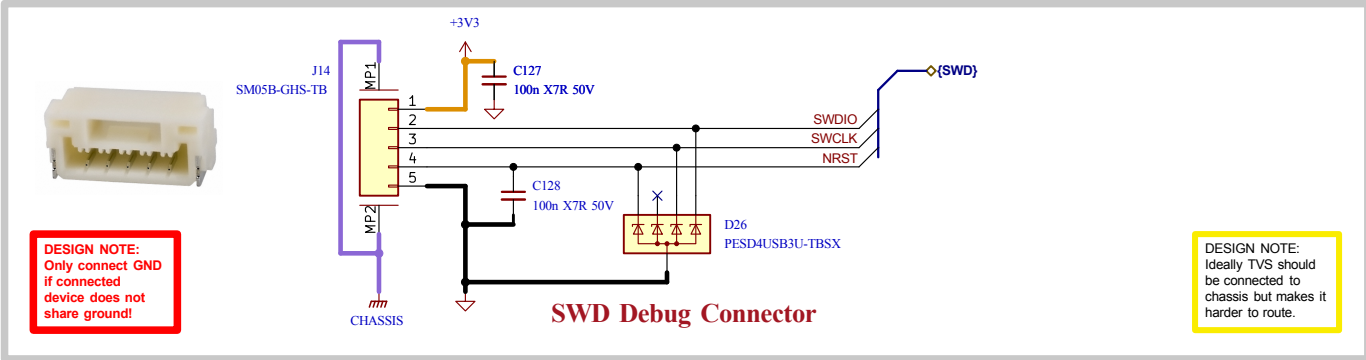
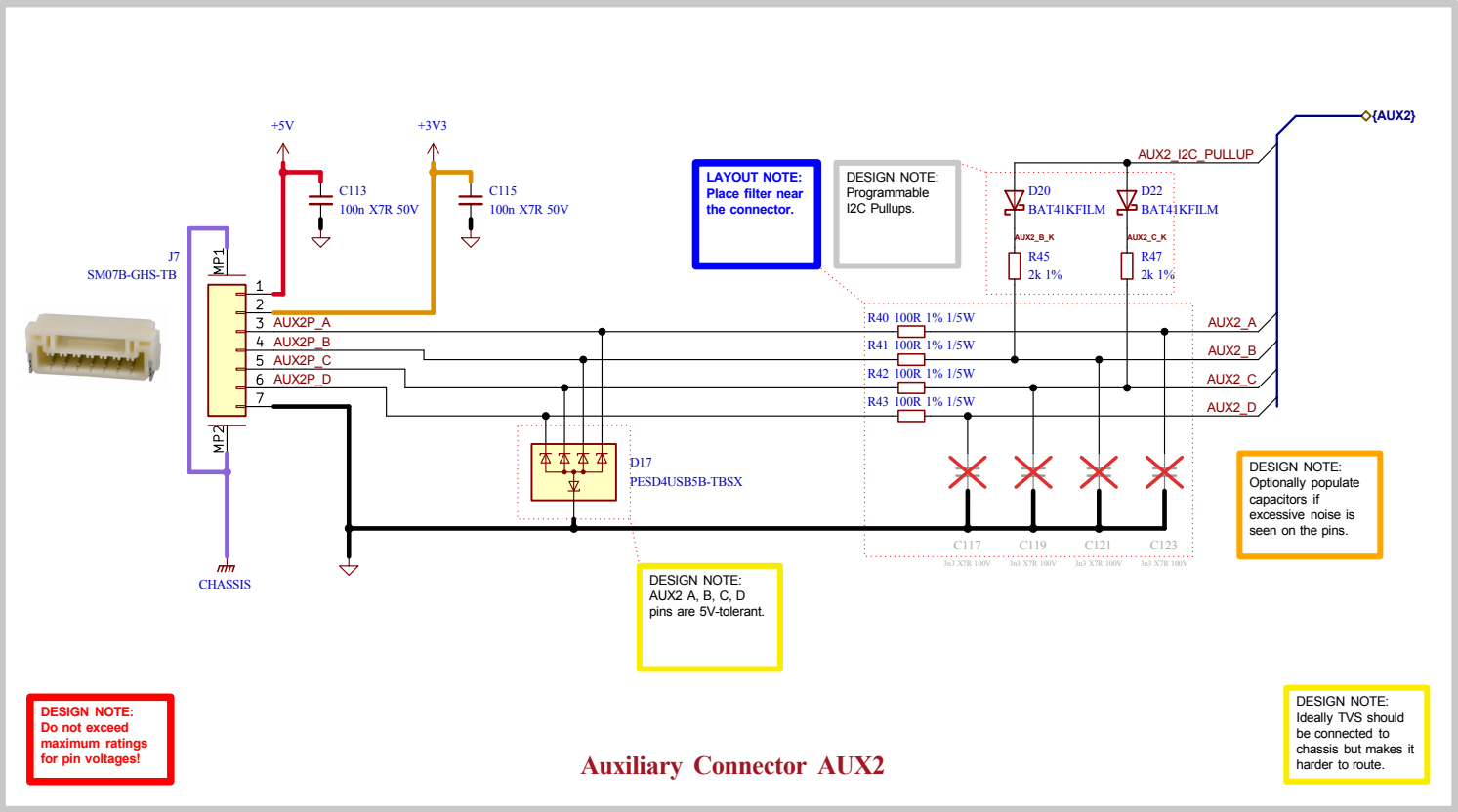
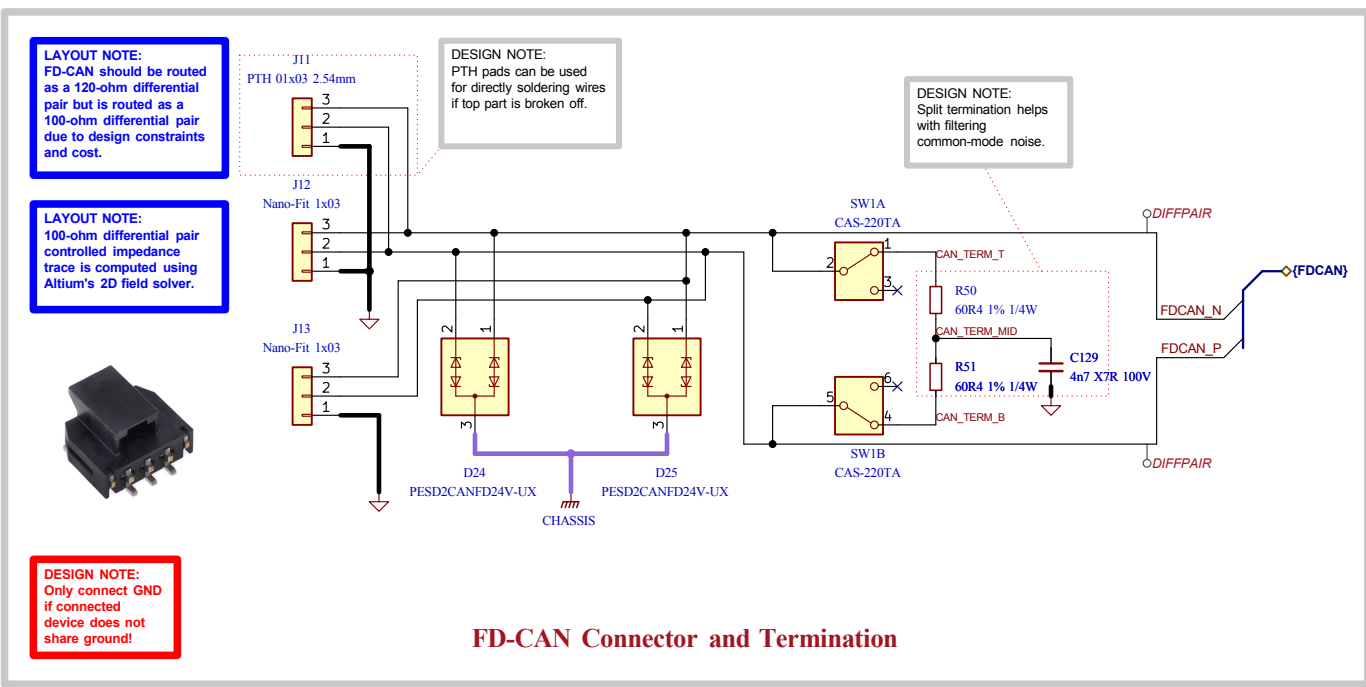
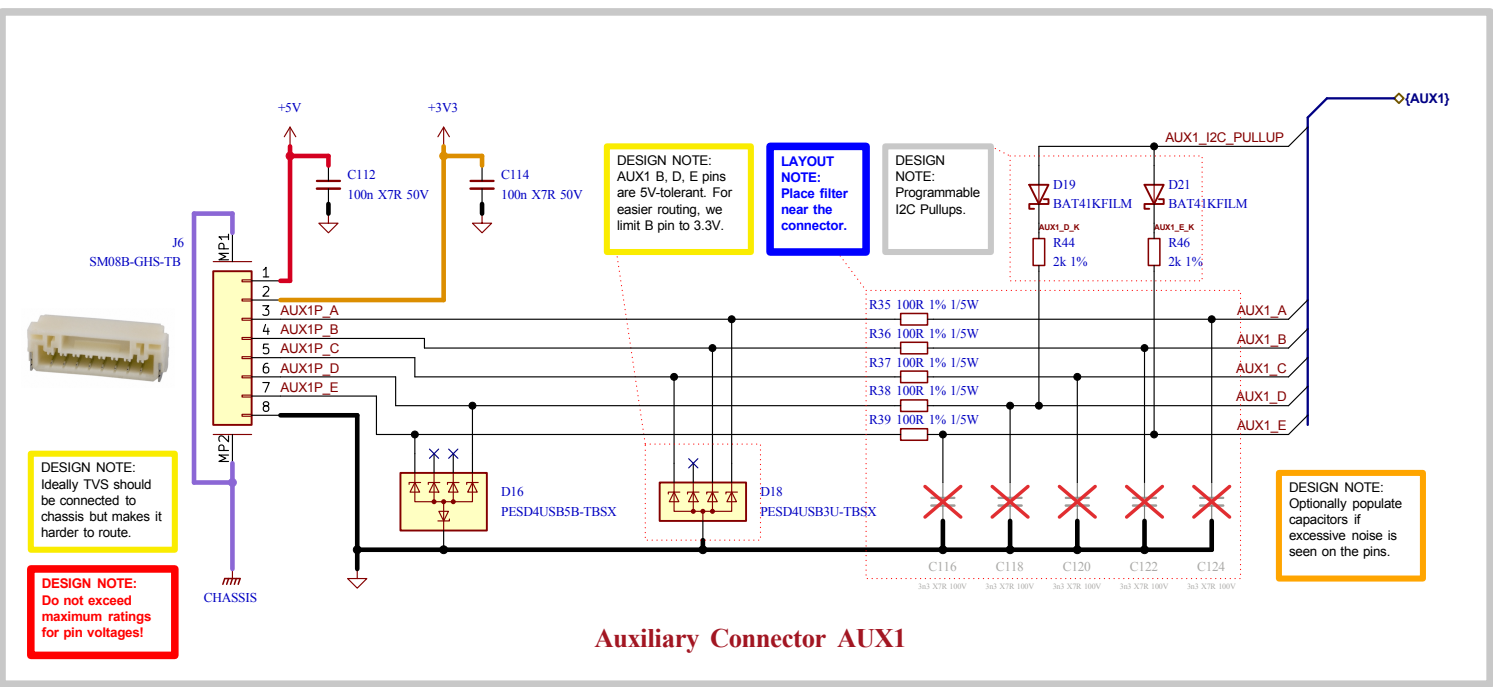
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|  |                                                          | Board Name:<br>Amulet Motion Controller    |                             | Project Name:<br>Chienpanzé |                                  |
|  | Sheet Title:<br>Interface - FD-CAN                       | File Name:<br>Interface - FD-CAN.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-10-15         | Revision:<br>1.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/Interface - FD-CAN/ |                                            | Reviewer:                   | Size:<br>A4                 | Sheet:<br>16 of 21               |

[17] Interface - Fan Control



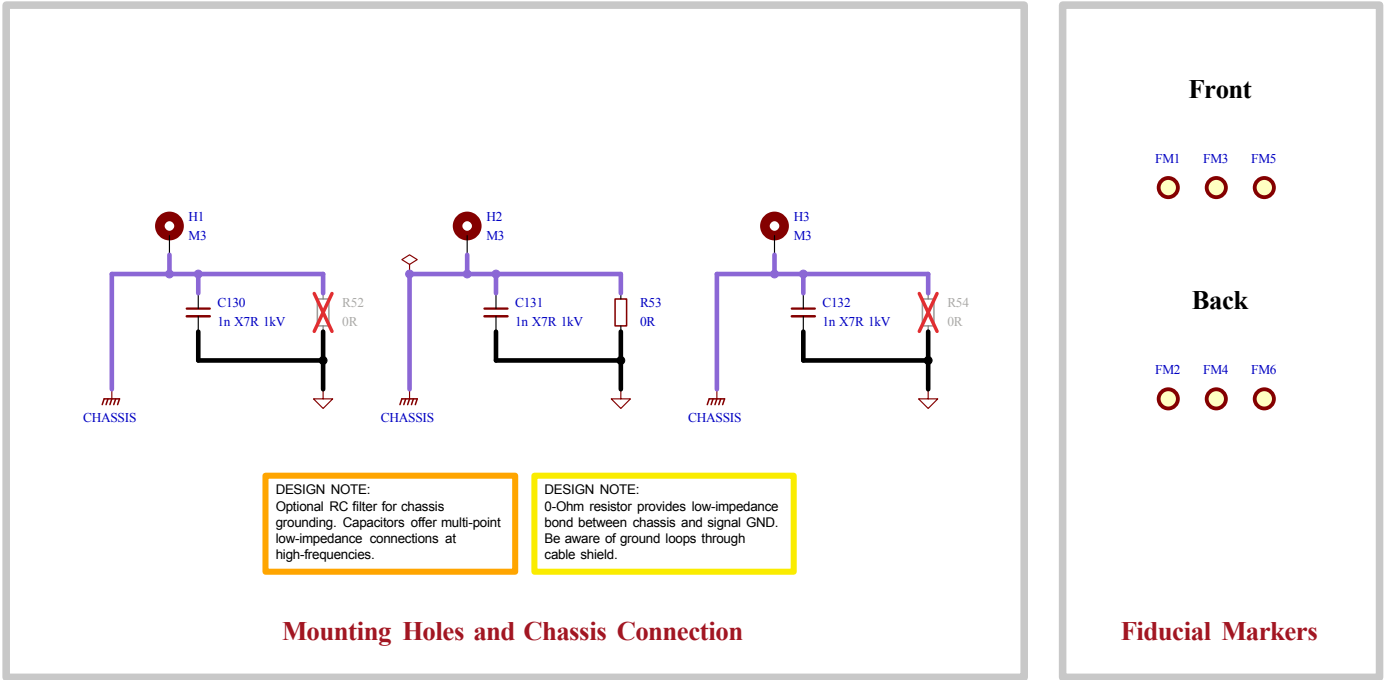
|  |                                                               |                                                 |                             |                             |                                  |
|--|---------------------------------------------------------------|-------------------------------------------------|-----------------------------|-----------------------------|----------------------------------|
|  | Comments:                                                     | Company:<br>EPFL Xplore Research                |                             | Variant:                    | Git Hash:<br>5b68472             |
|  |                                                               | Board Name:<br>Amulet Motion Controller         |                             | Project Name:<br>Chienpanzé |                                  |
|  | Sheet Title:<br>Interface - Fan Control                       | File Name:<br>Interface - Fan Control.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-11-19         | Revision:<br>1.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/Interface - Fan Control/ |                                                 | Reviewer:                   | Size:<br>A4                 | Sheet:<br>17 of 21               |

# [18] Interface - Interconnects



|  |                                                                                    |                                                                      |                                                |                                        |                                                     |
|--|------------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------|----------------------------------------|-----------------------------------------------------|
|  | <div>Comments:</div> <div>Reference:</div> <div>Flexible I/O worked examples</div> | <div>Company:</div> <div>EPFL Xplore Research</div>                  | <div>Variant:</div>                            |                                        | <div>Git Hash:</div> <div>5b68472</div>             |
|  |                                                                                    | <div>Board Name:</div> <div>Amulet Motion Controller</div>           | <div>Project Name:</div> <div>Chienpanzé</div> |                                        |                                                     |
|  | <div>Sheet Title:</div> <div>Interface - Interconnects</div>                       | <div>File Name:</div> <div>Interface - Interconnects.kicad_sch</div> | <div>Designer:</div> <div>Vincent Nguyen</div> | <div>Date:</div> <div>2023-12-31</div> | <div>Revision:</div> <div>1.1.1+ (Unreleased)</div> |
|  | <div>Sheet Path:</div> <div>/Project Architecture/Interface - Interconnects/</div> |                                                                      | <div>Reviewer:</div>                           | <div>Size:</div> <div>A3</div>         | <div>Sheet:</div> <div>18 of 21</div>               |

# [19] Misc - Holes, Fiducials



|  |                                                               |                                                |                                         |                     |                                  |                      |
|--|---------------------------------------------------------------|------------------------------------------------|-----------------------------------------|---------------------|----------------------------------|----------------------|
|  | Comments:                                                     |                                                | Company:<br>EPFL Xplore Research        |                     | Variant:                         | Git Hash:<br>5b68472 |
|  |                                                               |                                                | Board Name:<br>Amulet Motion Controller |                     | Project Name:<br>Chienpanzé      |                      |
|  | Sheet Title:<br>Misc - Holes, Fiducials                       | File Name:<br>Misc - Holes Fiducials.kicad_sch | Designer:<br>Vincent Nguyen             | Date:<br>2023-10-22 | Revision:<br>1.1.1+ (Unreleased) |                      |
|  | Sheet Path:<br>/Project Architecture/Misc - Holes, Fiducials/ |                                                | Reviewer:                               | Size:<br>A4         | Sheet:<br>19 of 21               |                      |

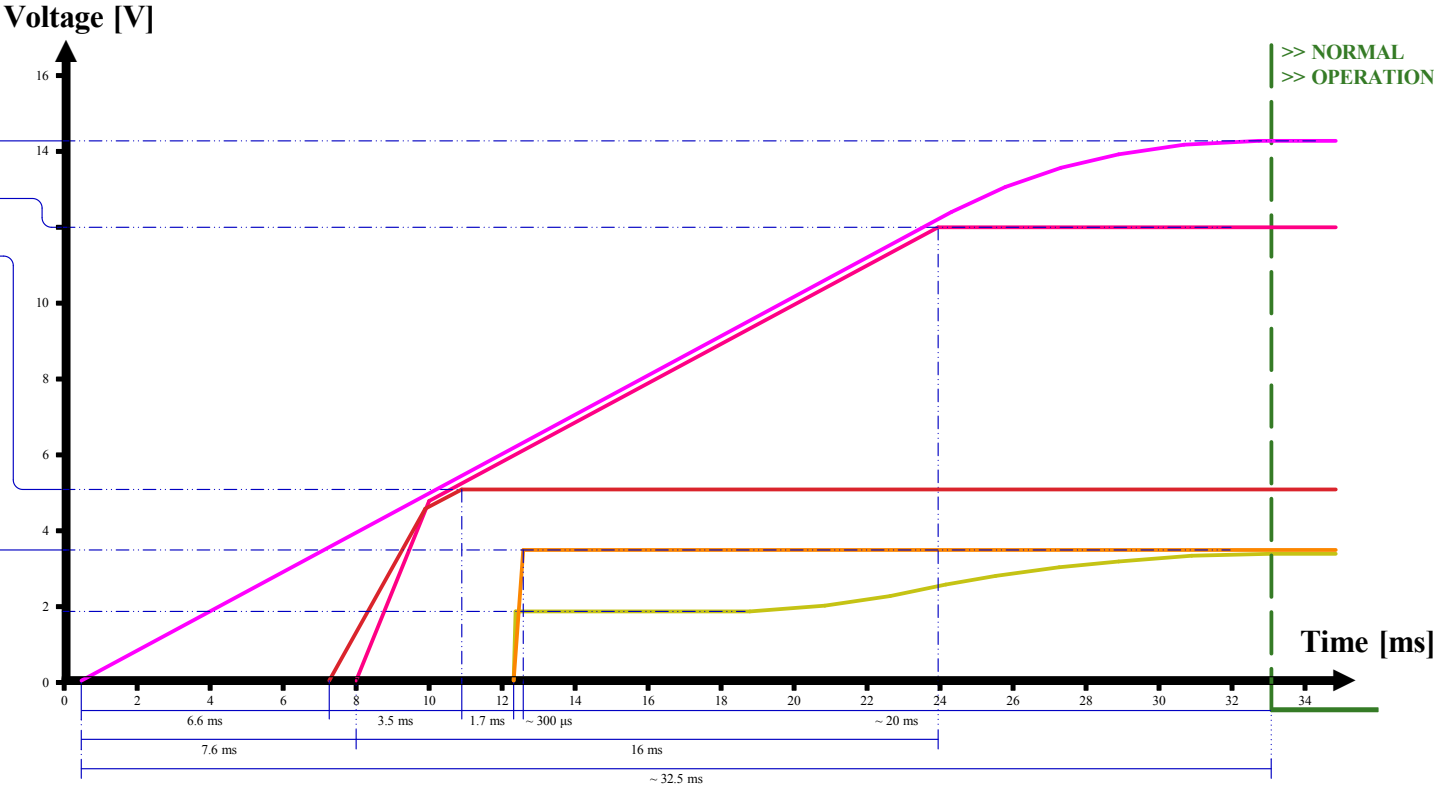


# [20] Power - Sequencing

| NAME  | SOURCE   | LEVEL       |
|-------|----------|-------------|
| +VBAT | BATTERY  | 12 - 44V    |
| +12V  | LMR36006 | 12V ± 1.5%  |
| +5V   | LMR36506 | 5V ± 1.5%   |
| +3V3  | TPS62172 | 3.3V ± 3%   |
| +A3V3 | LP2992   | 3.3V ± 0.5% |

DESIGN NOTE:  
Power sequencing graph measured experimentally.

DESIGN NOTE:  
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



|  |                                     |                                            |                             |                                  |
|--|-------------------------------------|--------------------------------------------|-----------------------------|----------------------------------|
|  | Comments:                           | Company:<br>EPFL Xplore Research           | Variant:                    | Git Hash:<br>5b68472             |
|  |                                     | Board Name:<br>Amulet Motion Controller    | Project Name:<br>Chienpanzé |                                  |
|  | Sheet Title:<br>Power - Sequencing  | File Name:<br>Power - Sequencing.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-03-12              |
|  | Sheet Path:<br>/Power - Sequencing/ | Reviewer:                                  | Size:<br>A4                 | Revision:<br>1.1.1+ (Unreleased) |
|  |                                     |                                            | Sheet:<br>20 of 21          |                                  |

# [21] Revision History

## Version 1.0.0 - 2023-12-12

- Added
- TVS protection and termination switch to FD-CAN.
  - Low-side switched 12V 600 mA source for external fan.
  - LDO for analog supply.
  - One TVS diode per power connector.
  - Second on-board I2C magnetic encoder for disambiguation.
  - ESD protection to all interfaces.
  - Over-voltage protection on thermistor ADC inputs.
  - Pi filters to inputs of buck regulators and MCU analog supply.
  - Decoupling caps next to power pins of connectors.
- Changed
- CPH-CPL capacitor to 47 nF (gate driver).
  - FD-CAN transceiver IC.
  - FETs for top cooled variant.
  - Input power TVS diode to bidirectional.
  - Moved SOx low-pass filter to MCU section.
  - PWM\_PHASEA with PWM\_PHASEC on STM32G474 pinout for easier routing.
  - RS422 pinout on connector.
  - Buck regulators to optimize for low noise.

## Version 1.0.1 - 2024-01-25

- Added
- Controller target specifications.
  - Credits to moteus on cover page.
  - Optional RC-Snubber to power stage.
- Fixed
- Chassis guard ring to go around the board.
  - CAN and power TVS diodes now go to chassis.
  - Clearance between nets to respect IEC60664-1 where possible.
  - Comment on precharge.
  - Power TVS diode reference designator from "U" to "D".
- Changed
- 5V 300 mA buck converter with 600 mA version.
  - Chassis-GND capacitor by 1nF 1kV.

## Version 1.0.2 - 2024-03-12

- Changed
- Power sequencing graph according to experimental data.

## Version 1.1.0 - 2024-04-13

- Added
- RC snubber passive values.
- Fixed
- More vias for VBUS and LMR36006 GND pads.
  - Board version voltage reference from +3V3 to +A3V3.
- Changed
- Motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

## Version 1.1.1 - 2025-01-18

- Fixed
- Replace non-stocked components.

|  |                                   |                                          |                             |                             |                                  |
|--|-----------------------------------|------------------------------------------|-----------------------------|-----------------------------|----------------------------------|
|  | Comments:                         | Company:<br>EPFL Xplore Research         |                             | Variant:                    | Git Hash:<br>5b68472             |
|  |                                   | Board Name:<br>Amulet Motion Controller  |                             | Project Name:<br>Chienpanzé |                                  |
|  | Sheet Title:<br>Revision History  | File Name:<br>Revision History.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-01-03         | Revision:<br>1.1.1+ (Unreleased) |
|  | Sheet Path:<br>/Revision History/ |                                          | Reviewer:                   | Size:<br>A4                 | Sheet:<br>21 of 21               |